

CHAPTER 4

CURVED BEAMS

INTRODUCTION

- Our analysis of stresses due to bending has been restricted so far to **Straight members**.
- The flexure formula applies to a straight member, the normal strain within it varies linearly from the neutral axis. If the member is curved, however, this assumption becomes inaccurate, and so we must develop another method to describe the stress distribution.
- **A Curved Beam** is a member in which the neutral axis in unloaded condition is curved instead of straight.

- In all the cases, the members are not slender, but rather have a sharp curve, and their cross-sectional dimensions are large compared with their radius of curvature.

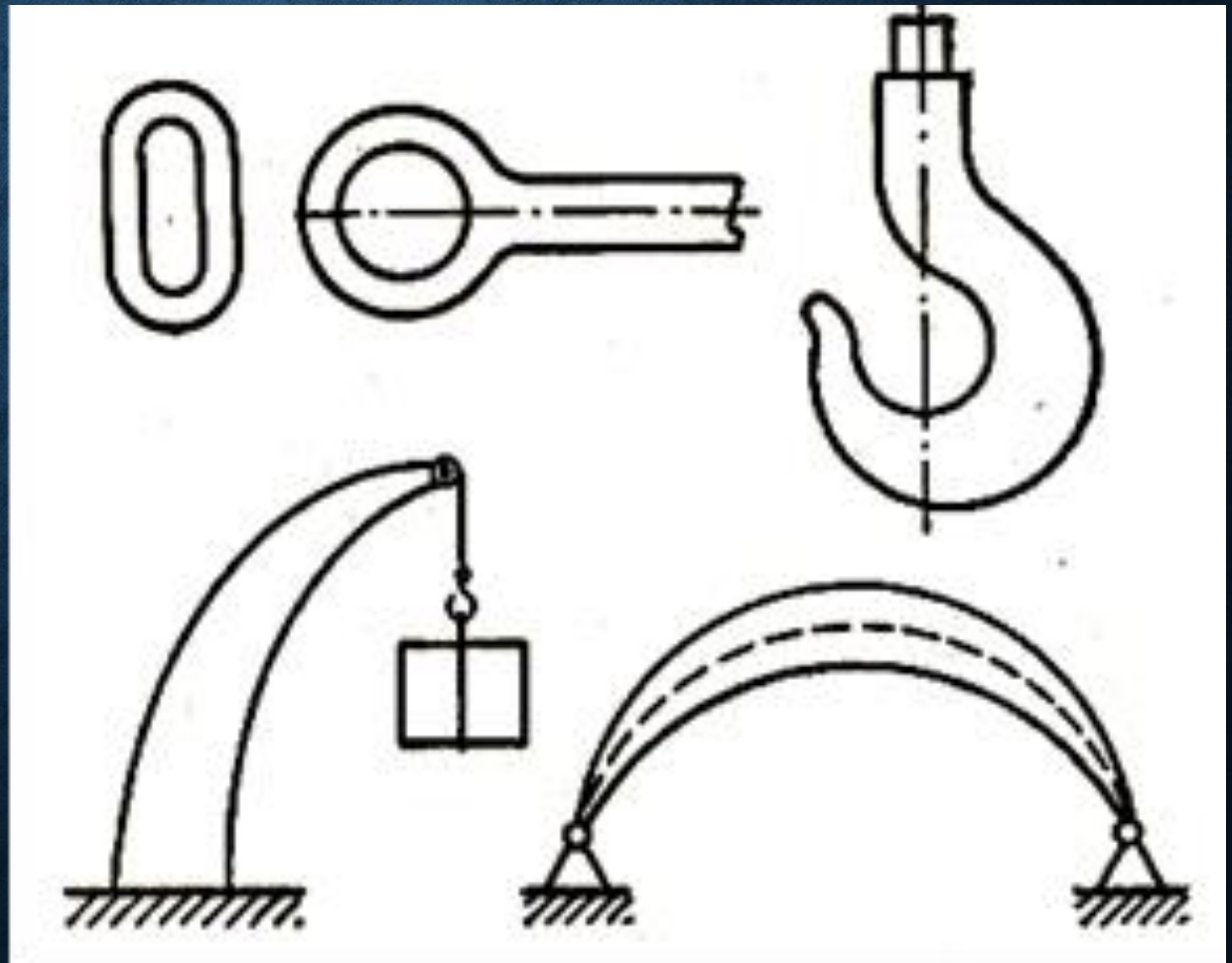


Figure 4.1 Examples
Of curved beams

- Consider the curved member of uniform cross section shown in Figure 4.2. Its transverse section is symmetric with respect to the y axis, in its unstressed state, its upper and lower surfaces intersect the vertical xy plane along arcs of circle AB and FG centered at C.

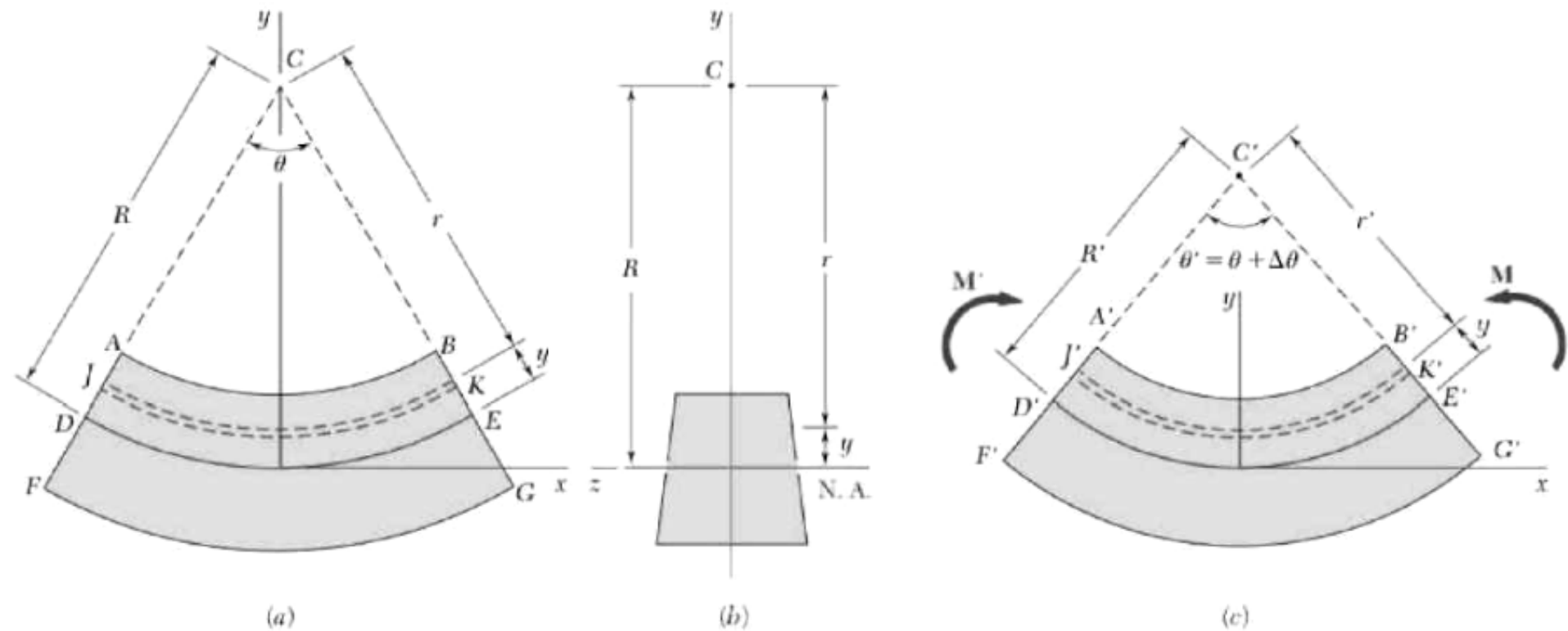


Figure 4.2: Curved member

- We now apply two equal and opposite couples M and M' in the plane of symmetry of the member (Fig.4.2 C). If the couples M and M' are directed as shown, the curvature of the various arcs of circle will increase; that is $A'C' < AC$.
- We also note that the couples M and M' will cause the length of the upper surface of the member to decrease ($A'B' < AB$) and the length of the lower surface to increase ($F'G' > FG$).
- We conclude that a neutral axis must exist in the neutral surface with the xy plane as represented in Fig.4.2(a) by the arc DE of radius R , and in Fig.4.2(C) by the arc $D'E'$ of radius R' .

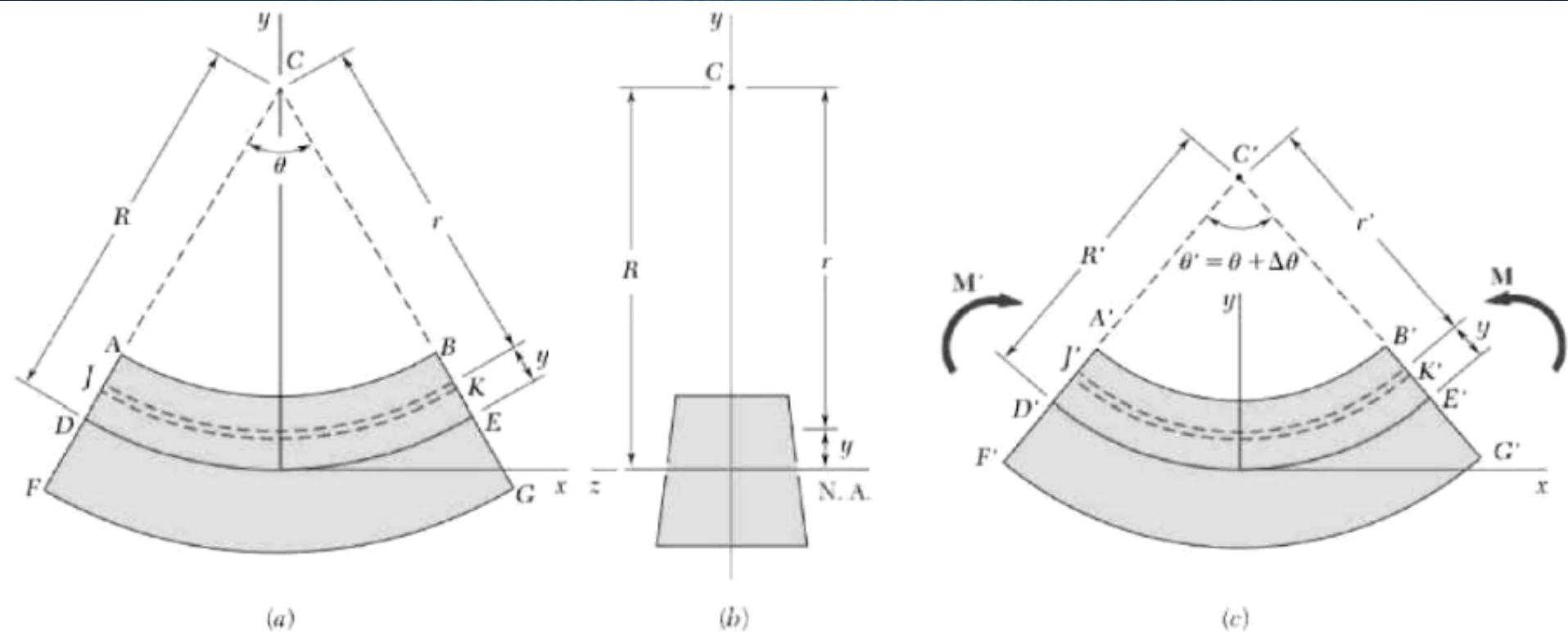


Figure 4.2: Curved member

- Denoting by θ and θ' the central angles corresponding respectively to DE and D'E'. We express the fact that the length of the neutral surface remains constant by writing

$$R\theta = R'\theta' \dots\dots\dots(1)$$

- Considering now the arc of circle JK located at a distance y above the neutral surface, and denoting respectively by r and r' the radius of this arc before and after the bending couples have been applied, we express the deformation of JK as

$$\delta = r'\theta' - r\theta \dots\dots\dots(2)$$

- Observing from fig.4.2 that

$$r = R - y \quad r' = R' - y \dots \dots \dots (3)$$

- And substituting these expressions in to Eq(2), we write

$$\delta = (R' - y)\theta' - (R - y)\theta \quad \theta' - \theta = \Delta\theta$$

Or, recalling Eq.(1) and setting

$$\delta = -y\Delta\theta \dots \dots \dots (4)$$

- The normal strain ϵ_x in the elements of JK is obtained by dividing the deformation δ by the original length $r\theta$ of arc JK.

We write

$$\epsilon_x = \frac{\delta}{r\theta} = -\frac{y\Delta\theta}{r\theta}$$

- Or, recalling the first of the relation Eq.(3),

$$\epsilon_x = -\frac{\Delta\theta}{\theta} \frac{y}{R-y} \dots\dots\dots(5)$$

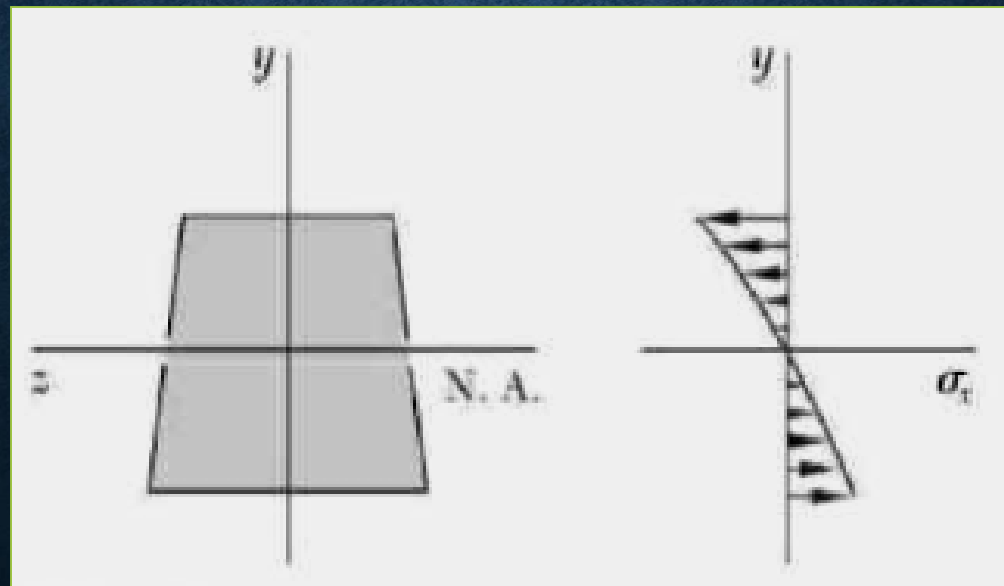
- The relation obtained shows that, while each transverse section remains plane, the normal strain ϵ_x does not vary linearly with the distance y from the neutral surface.
- The normal stress σ_x can now be obtained from Hooke's law, $\sigma_x = E\epsilon_x$ by substituting for ϵ_x from Eq.(5). We have

$$\sigma_x = -\frac{E\Delta\theta}{\theta} \frac{y}{R-y} \dots\dots\dots(6)$$

- Or, alternatively, recalling Eq.(3),

$$\sigma_x = -\frac{E\Delta\theta}{\theta} \frac{R-r}{r} \dots\dots\dots(7)$$

- Equation (6) shows that, like ϵ_x , the normal stress σ_x does not vary linearly with the distance y from the neutral surface. Plotting σ_x versus y , we obtain an arc of **hyperbola** (Fig.4.3).



• Figure 4.3

- In order to determine the location of the neutral surface in the member and the value of the coefficient $E\Delta\theta/\theta$ used in Eqs.(6) and (7), we now recall that the elementary forces acting on any transverse section must be statically equivalent to the bending couple M .
- The sum of the elementary forces acting on the section **must be zero**, and that the sum of their moments about the transverse z axis must be equal to the bending moment M , we write the equations

$$\int \sigma_x dA = 0 \dots\dots\dots(i)$$

and

$$\int (-y \sigma_x dA) = M \dots\dots\dots(ii)$$

- Substituting for σ_x from (7) in to Eq(i), we write

$$-\int \frac{E \Delta \theta}{\theta} \frac{R-r}{r} dA = 0$$

$$\int \frac{R-r}{r} dA = 0$$

$$R \int \frac{dA}{r} - \int dA = 0$$

- The distance R from the center of curvature C to the neutral surface is defined by the relation.

$$R = \frac{A}{\int \frac{dA}{r}} \dots\dots\dots(8)$$

- The value obtained for R is not equal to the distance from C to the centroid of the cross section, since $\bar{r}$ is defined by a different relation, namely,

$$\bar{r} = \frac{1}{A} \int r dA \dots\dots\dots(9)$$

Conclusion

- **In a curved member, the neutral axis of a transverse section does not pass through the centroid of that section.**

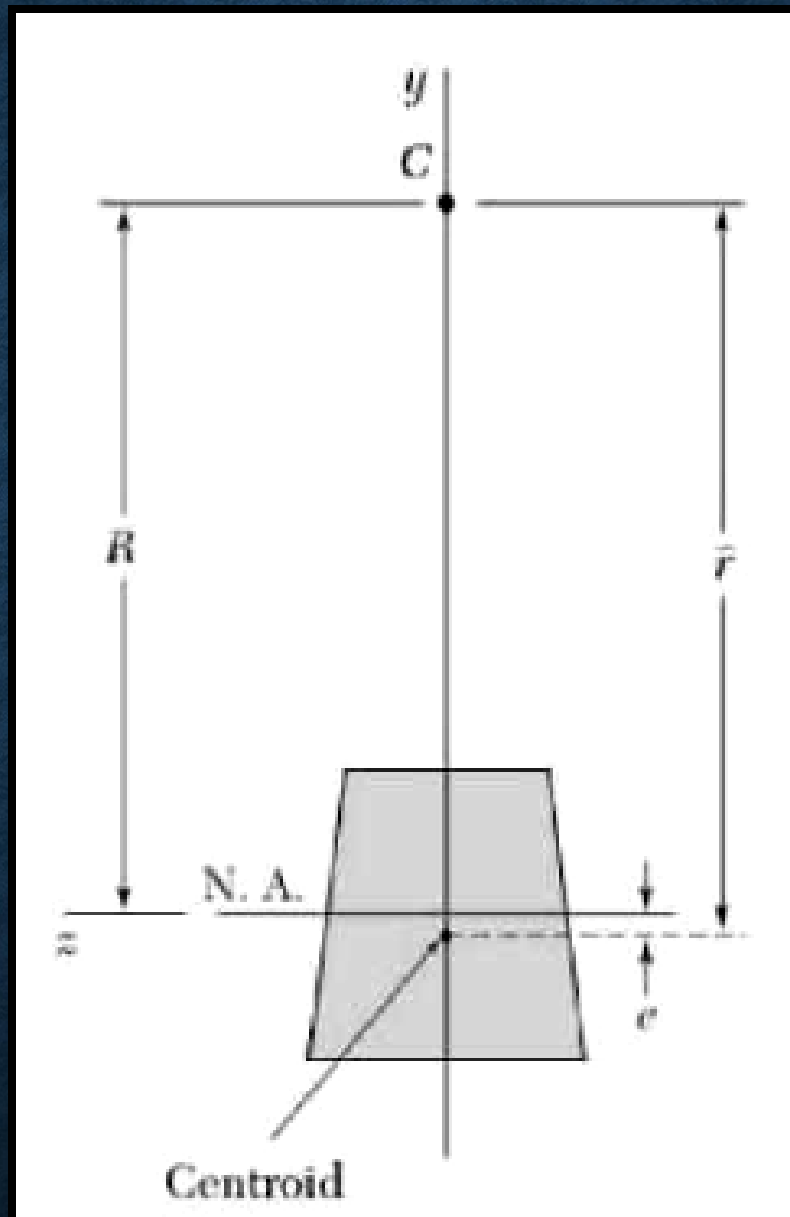
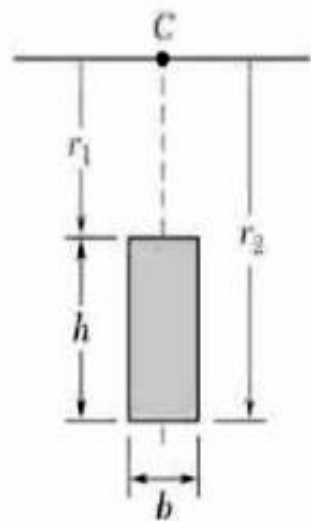
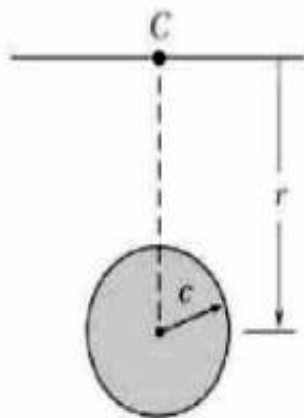


Figure 4.4



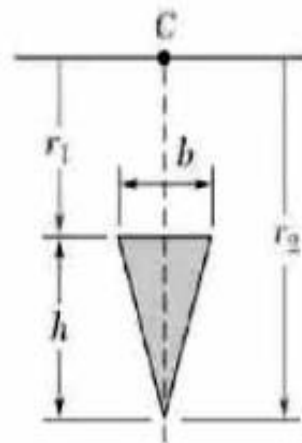
Rectangle

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$



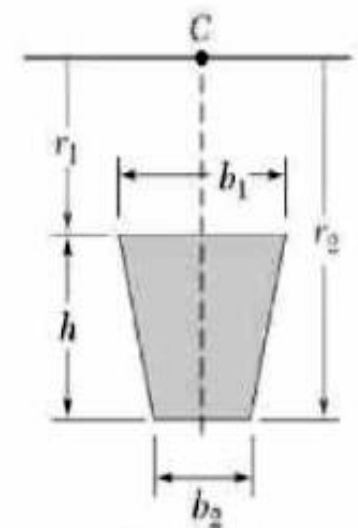
Circle

$$R = \frac{1}{2} (\bar{r} + \sqrt{\bar{r}^2 - c^2})$$



Triangle

$$R = \frac{\frac{1}{2}h}{\frac{r_2}{h} \ln \frac{r_2}{r_1} - 1}$$



Trapezoid

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1) \ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

Figure 4.5: Radius of neutral surface for various cross-sectional shapes

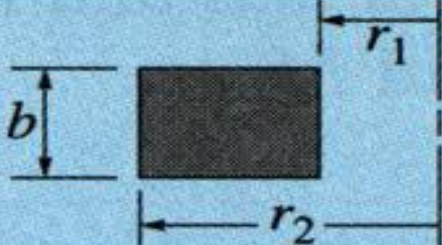
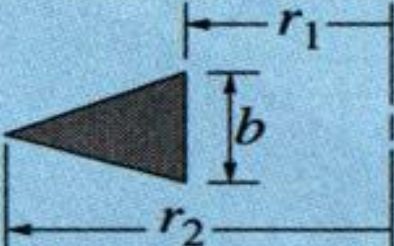
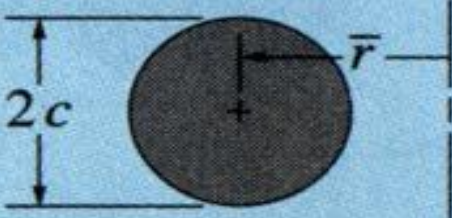
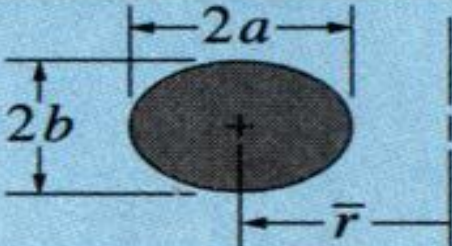
Shape	Area	$\int_A \frac{dA}{r}$
	$b(r_2 - r_1)$	$b \ln \frac{r_2}{r_1}$
	$\frac{b}{2}(r_2 - r_1)$	$\frac{b}{2} \frac{r_2}{(r_2 - r_1)} \left(\ln \frac{r_2}{r_1} \right) - b$
	πc^2	$2\pi \left(\bar{r} - \sqrt{\bar{r}^2 - c^2} \right)$
	πab	$\frac{2\pi b}{a} \left(\bar{r} - \sqrt{\bar{r}^2 - a^2} \right)$

Table 4.6 Areas and Integrals for Various Cross Sections

- Substituting now for σ_x , from Eq.(7) in to Eq(ii), we write

$$\int \frac{E\Delta\theta}{\theta} \frac{R-r}{r} y dA = M$$

since $y = R - r$

$$\frac{E\Delta\theta}{\theta} \int \frac{(R-r)^2}{r} dA = M$$

- Expanding the square in the integrand, we obtain after reductions

$$\frac{E\Delta\theta}{\theta} \left[R^2 \int \frac{dA}{r} - 2RA + \int r dA \right] = M$$

- Recalling Eqs.(8) and (9), we note that the first term in the brackets is equal to $\mathbf{R}\bar{\mathbf{A}}$, while the last term is equal to $\bar{r}A$. We have, therefore,

$$\frac{E\Delta\theta}{\theta} \left(RA - 2RA + \bar{r}A \right) = M$$

and, solving for $E\Delta\theta/\theta$,

$$\frac{E\Delta\theta}{\theta} = \frac{M}{A(\bar{r} - R)} \dots\dots\dots(10)$$

- Referring to Fig.(4.2), we note that $\Delta\theta > 0$ for $M > 0$. It follows that , regardless of the shape of the section.

$$\bar{r} - R > 0 \text{ or } R < \bar{r}$$

- Thus, the neutral axis of a transverse section is always located between the centroid of the section and the center of curvature of the member (Fig 4.4). Setting $\bar{r} - R = e$, we write Eq.(10) in the form

$$\frac{E\Delta\theta}{\theta} = \frac{M}{Ae} \dots\dots\dots(11)$$

- Substituting now for $E\Delta\theta/\theta$ from (11) in to Eqs.(7) and (8), we obtain the following alternative expressions for the normal stress in a curved beam:

$$\sigma_x = -\frac{My}{Ae(R-y)} = \frac{M(r-R)}{Aer} \dots\dots\dots(12)$$

END